Do Now · retrieval

C# Login Project · Code a Hash Table

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

A login check uses validUser && validPassword . What is the result when validUser is true and validPassword is false? Explain.

Q2

A D-type flip-flop currently stores Q = 1. D changes to 0 but there is no triggering clock edge. What is Q?

Q3

A queue receives A, B and C in that order. Remove one item, then enqueue D. List the remaining items from front to rear.

Q4

A five-bucket hash table uses h(k) = k MOD 5 with linear probing. Insert keys 7, 12 and 17 in that order into an empty table. State each final index.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

The problem: more than one user — A single username comparison is not scalable. We need a way to store and find many usernames.

Core idea 2

What is a hash function? — Convert the username to a number, then use % tableSize to choose a bucket index. Our small function is deliberately simple for learning.

Core idea 3

What if two names choose one bucket? — That is a collision. Use linear probing: try the next bucket, wrapping to the beginning when necessary.

Key visual · complete concept



"ab" and "ba" both start at index 6

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Hash-table lookup versus password hashing — The hash-table function locates an account. It does NOT securely protect a password. Use invented demo passwords only.

Core idea 5

MCQ revision — Write four precise MCQs on linked lists, stacks, queues and trees; extend with BST and hash-table MCQs if time allows.

Vocabulary · use the correct technical terms

A table using a hash function to locate data from a key. — A table using a hash function to locate data from a key.

A repeatable mapping from key to bucket index. — A repeatable mapping from key to bucket index.

A location in the hash table. — A location in the hash table.

Two keys map to the same bucket. — Two keys map to the same bucket.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Build a real hash-table index

Follow the worked method

1. Create the buckets — CODE:

```
const int SIZE = 7;
string?[] usernames = new string?[SIZE];
```

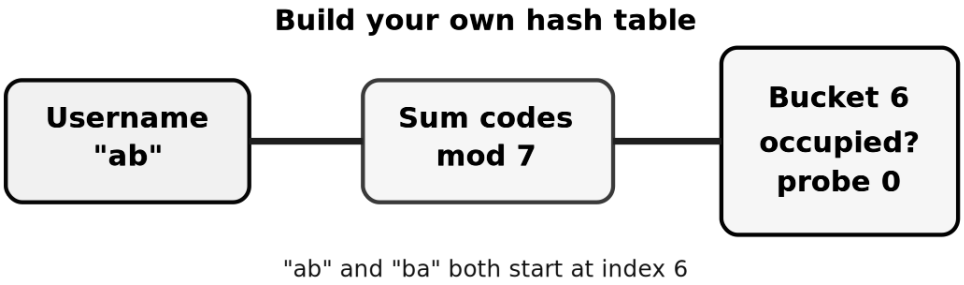
2. Write the index function — CODE:

```
static int Hash(string key, int size)
{
int total = 0;
foreach (char letter in key)
total += (int)letter;
return total % size;
}
```

3. Explain the limitation — CODE:

```
// Hash("ab", 7) == Hash("ba", 7)
// Same total -> collision.
// This is an educational hash function,
// NOT a secure password hash.
```

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain your solution first
What happens when a bucket is already occupied?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

CODING 1 · INSERT

Create a seven-bucket string array, implement Hash(), and insert at least four test usernames. Print each chosen bucket index.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

SYNOPTIC MCQ BANK

Write MCQs on linked lists, directed/undirected graphs, stacks, queues, trees, binary search trees and hash tables. Make a six-question draft; each needs four options and a written explanation.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A hash-table function based on adding character codes is suitable for protecting real passwords.

Use key vocabulary from memory

1. Explain A table using a hash function to locate data from a key. in your own words.

2. Explain A repeatable mapping from key to bucket index. in your own words.

3. Explain A location in the hash table. in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.